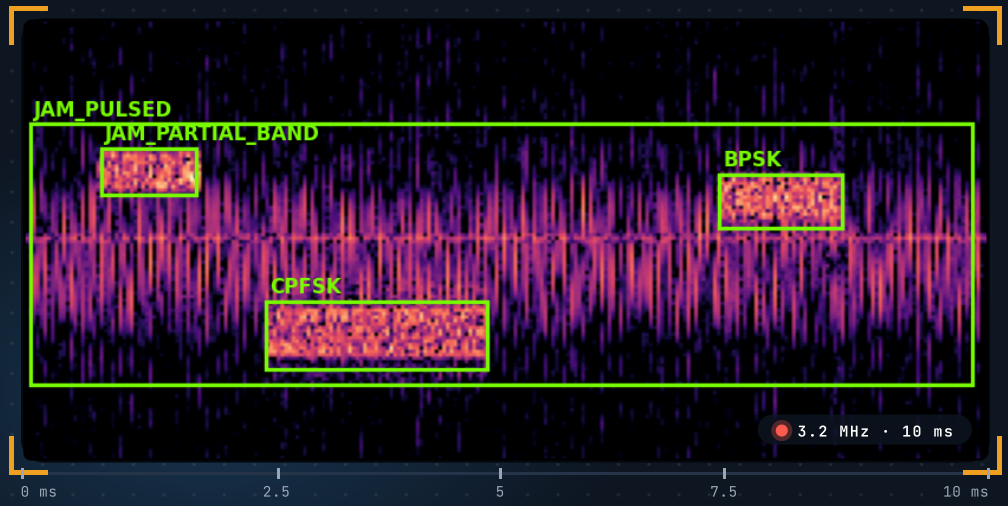
NEXA

SEDIC@FOI2026



NEXA's own boxes and answers on a simulated capture at 20 dB, not retouched: a pulsed jammer laid over three other signals, and every one still found and named.

TRACK I · RF SIGNAL INTELLIGENCE

Every emitter on the band, named.

NEXA watches a whole band of radio spectrum, finds every transmission in it and tells the operator what each one is: jammer, radar, voice or data. It runs offline on one workstation, and people stay in control.

60	**90%**	**79.4%**	**0**
signal classes in 12 families	right signal family, 15 dB, 8-window vote	exact class out of 60, SNR ≥ 10 dB, 16-window vote	data leaves the machine

02 THE PROBLEM

One band. Many emitters. All at once.

In contested spectrum, friendly links, radars and deliberate interference overlap in time and frequency. An operator has seconds to work out what is on the air. NEXA answers three questions for every emitter it finds.

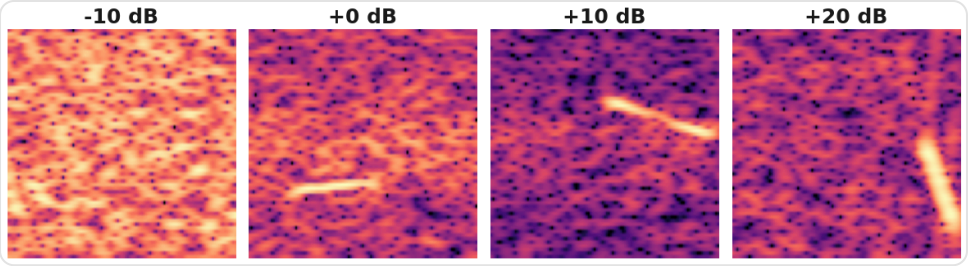
WHAT
What is it?
One of 60 signal types, or the family when the exact type cannot be told apart.

HOW SURE
How sure is it?
A confidence band on every answer: high, medium or uncertain.

WHERE
Where and when?
A box on the spectrogram: its frequency, start time and duration.

WHAT THE OPERATOR SEES · ONE ROW PER EMITTER		
`RADAR_PULSE + JAM_DRFM`	A jammer copying a radar: both on the air	HIGH
`QPSK`	A digital data link	HIGH
`QAM 16–256`	Sizes look alike: likeliest 64QAM	MEDIUM
`weak burst`	Too weak to name: flagged for a person	UNCERTAIN

Two answers can both be true. Each class is its own yes or no, so a radar and the jammer sitting on top of it are reported together, not as one or the other.



The same radar chirp from −10 dB (buried in noise) to +20 dB (clear). NEXA is trained across the whole range, so it learns when to say “uncertain”.

03 PERFORMANCE

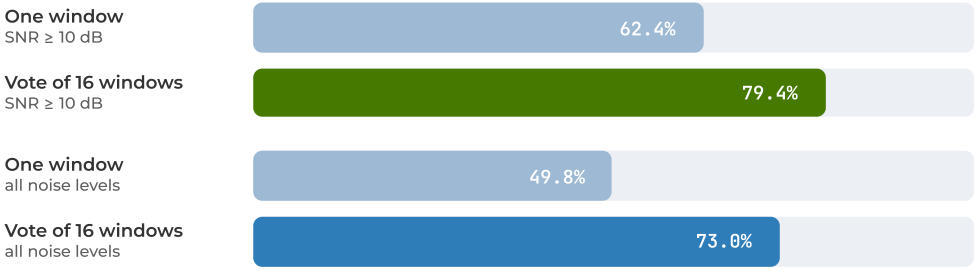
Named exactly, out of 60.

79.4% exact class, per emitter

All 60 classes · SNR ≥ 10 dB · vote over 16 windows (2.6 ms of signal) · fixed test set of 120,000 windows the model never trained on. Chance is 1.7%.

From one window to a vote

One window is 160 µs. The tracker averages every window of the same emitter, and the answer firms up.



Exact class out of 60, same fixed test set. “All noise levels” includes signals buried at −10 dB.

It says how sure it is

HIGH MEDIUM UNCERTAIN

Confidence is calibrated on held-out data, and every answer carries a band. Uncertain answers go to a person, not into the log as fact.

It does not guess what it cannot see

QAM, APSK and high-order PSK sizes look alike in 160 µs. NEXA names the group and offers the likeliest size as a hint. An empty channel is reported as empty.

Right family: 90% across all 12

ASK	100%	FSK	100%
Radar	95%	QAM	94%
Jammers	93%	OFDM / LTE / NR	93%
PSK	91%	Analog voice	86%
Spread spectrum	82%	Pulse comms	70%
APSK	69%	Noise floor	66%

Answer lands in the right family · 15 dB · vote over 8 windows · 32 fresh tracks per class from our own generators, one emitter at a time (cleaner than the test set) · physics_aug_best.

04 CLASS BY CLASS

60 signal classes, every one scored.

37 / 60 classes named exactly 90% of the time or more

48 / 60 classes named exactly at least half the time

10 / 10 jammer types recognised as jamming: family 93%

Families by tier: THREAT MILITARY CIVILIAN EMPTY

Jammers THREAT — family 93%
Barrage 100 · Spot 100 · Multitone 100 · Swept 100 · Chirp 100 · Pulsed 100 · Partial-band 100 · Reactive 100 · DRFM 81 · Tone 50

Radar MILITARY — family 95%
Pulse 100 · LFM 100 · Barker 100 · Frank 100 · Costas 100 · FMCW 100 · Poly-Barker 97 · CW 62

Spread spectrum MILITARY — family 82%
FHSS 100 · DSSS 94 · LoRa 53

Pulse comms MILITARY — family 70%
PPM 100 · SSR / IFF 100 · AFSK 9

Analog voice CIVILIAN — family 86%
AM-DSB-WC 100 · AM-DSB-SC 100 · AM-SSB-WC 100 · WBFM 100 · PM 94 · NBFM 50 · AM-SSB-SC 0

FSK CIVILIAN — family 100%
2FSK 100 · 4FSK 100 · 8FSK 100 · GFSK 100 · MSK 100 · CPFSK 100 · GMSK 78

ASK CIVILIAN — family 100%
OOK 100 · 4ASK 91 · 8ASK 84

OFDM / LTE / NR CIVILIAN — family 93%
NR-like 100 · LTE-like 78 · OFDM 12

PSK CIVILIAN — family 91%
OQPSK 100 · 16PSK 100 · QPSK 97 · BPSK 47 · 8PSK 0 · 32PSK 0
sizes answered as a group

QAM CIVILIAN — family 94%
256QAM 97 · 32QAM 47 · 16QAM 3 · 64QAM 0 · 128QAM 0
sizes answered as a group

APSK CIVILIAN — family 69%
32APSK 69 · 16APSK 66 · 64APSK 6 · 128APSK 0
sizes answered as a group

Noise floor EMPTY — family 66%
Nothing on air 66

90+ 50–89 <50 exact class, % · 15 dB · vote over 8 windows · 32 fresh tracks per class from our own generators, one emitter at a time · physics_aug_best. Sizes of QAM, APSK and high-order PSK look alike in 160 µs: NEXA answers the group and gives the likeliest size as a hint, so a low size score is still a right family answer.

05 HOW NEXA COMPARES

Built for contested spectrum, not just monitoring.

Commercial tools are mature and field-proven, and most are built to monitor and decode communications. NEXA is built for the case they do not describe: a jammer on top of a signal, radar and comms on one band, and an answer the operator can trust.

	NEXA this project	DeepSig OmniSIG AI RF sensing	R&S CA100 signal analysis software	PROCITEC go2MONITOR radio monitoring
What it names	60 classes: comms, 8 radar, 10 jammer types, noise	Signals the user trains it on	Modulation type and transmission system, HF–UHF	Modulation types and 350+ modems
How	Deep learning: three views of each window fused with physics features	Deep learning	Classifier with rule-based processing	Multi-stage classifier library
Jammer on top of a signal	Both reported (multi-label)	*not stated*	*not stated*	*not stated*
Confidence on each answer	Calibrated band: high, medium, uncertain	*not stated*	*not stated*	*not stated*
Voice and Morse to text	Speech and Morse to text, local AI summary	*not stated*	Demodulation, decoding, content recovery	Automatic decoding
Learns from operators	Two-person approval, gated retrain, rollback	Retrain by labelling captures (Studio)	*not stated*	User-built decoders
Runs on	One offline workstation, any SDR or IQ file	Edge devices and containers	PC, dongle licence	PC with supported receivers
Maturity	Research prototype	Commercial	Commercial	Commercial

Sees the attack
Ten jammer types and eight radar waveforms, named next to the signal they sit on.

Says how sure it is
Every answer carries a calibrated band, and uncertain ones go to a person.

Stays in the room
Detection, speech recognition and AI reading all run on one offline machine.

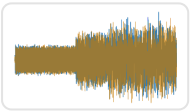
How we compared: from each vendor's public product pages, November 2026. “Not stated” means we found no public claim, not that the product cannot do it. Vendors do not publish accuracy on a shared test set, so we compare capabilities, not scores.

deepsig.ai/omnisig-sdk · rohde-schwarz.com (R&S CA100) · procitec.com (PRO Monitoring Suite tech. specs, products brochure 24.1)

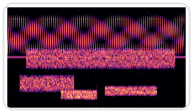
06 HOW IT WORKS

See the whole spectrum, then name every emitter.

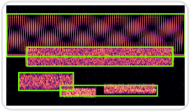
1 · SEE THE SPECTRUM



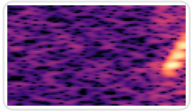
1 Capture
Raw IQ samples from a radio receiver, a recorded file or the simulator. Seconds of the whole band at once.



2 Spectrogram
The band as a picture: time across, frequency down. Every emitter leaves its own mark.

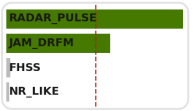


3 Detector
A YOLO model draws a box around every emitter in time and frequency, many at once.

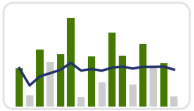


4 Cut and clean
Each box is filtered and cut into 160 µs windows at its own frequency; one per pulse for pulsed signals.

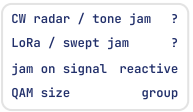
2 · NAME IT



5 Classifier
Tri-Branch Deep Fusion Network for Multi-Class RF Signal Classification: raw IQ, spectrogram and spectrum CNNs fused with physics features.



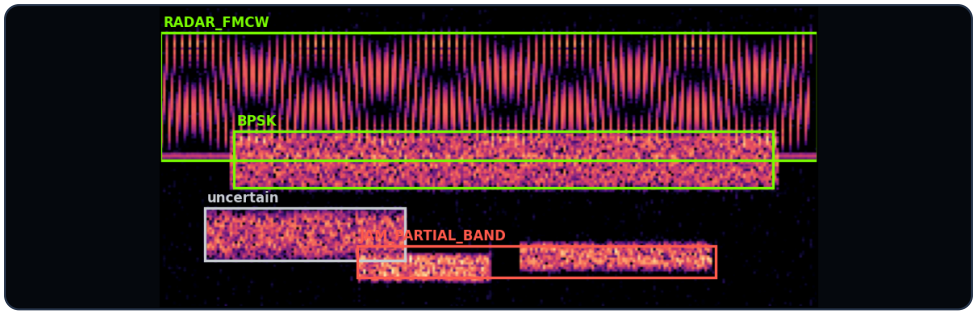
6 Tracker and vote
Joins one emitter's windows over time and averages them; a new signal in the band gets its own track.



7 Resolver
Settles look-alikes from behaviour over time, e.g. a jammer switching on over a signal is reactive.



8 Dashboard
Boxes on the spectrogram and one row per emitter: answer, confidence band, frequency, time, alerts.



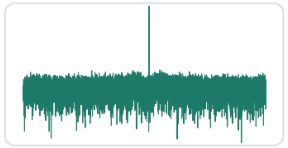
NEXA's own answers on a simulated capture at 15 dB, not retouched: **right**, **wrong**, **uncertain**. One 64QAM link was called a jammer; one jammer was missed.

NEXA

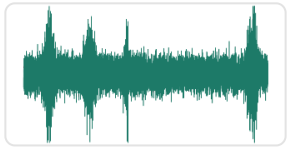
SEDIC 26 Strategic Electronic Defence Innovation Challenge 2026

07 LISTEN, READ, LEARN

Voice and Morse become text. People stay in control.



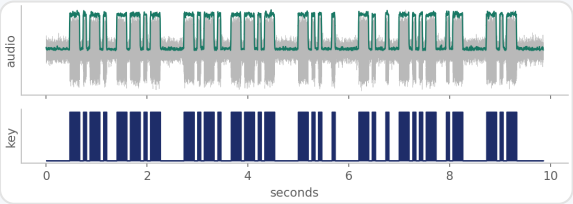
1 Zoom in
Measures carrier and sidebands; rules name it AM, FM, SSB or CW.



2 Demodulate
The right demodulator turns radio samples into audio.

speech
→ text
on this machine

3 Voice to text
Whisper speech recognition, run locally. No audio leaves.



CQ CQ DE NEXA K

Morse to text. Times each key press, measures the dot length, turns dots and dashes into letters (18 words per minute). A local AI summarises the text and anything not in the text is removed.

It learns, but only with approval

1 **Operator corrects**
Flags or fixes an answer with one click.

2 **Analyst approves**
A second person checks the stored signal.

3 **Retrain**
Started by a person, never on its own.

4 **Gate**
Goes live only if it beats the current model on the same test set; rollback any time.

HARDWARE
Receiver, file or simulator
A software-defined radio, a recorded IQ file or the built-in simulator, at any sample rate.

ACCOUNTABLE
Tamper-evident audit trail
Every capture, decision and correction is logged locally, each entry chained to the one before.

OFFLINE · TAMPER-EVIDENT AUDIT TRAIL · NO DATA LEAVES THE MACHINE

NEXA

AI-powered RF spectrum intelligence

60 signal classes · offline · human in the loop



Eavan Tan
GROUP LEADER

Eileen Yeoh Jing Han

Dr Noureen Talpur
ADVISOR

Jessy Pang Xin Yuan

Chua Xin Ying

SCAN TO SEE IT RUN

Try the console
In your browser

Watch the demo
NEXA running, on video

Keep this brochure
Scan to save or share

sedic-ai-nexa.vercel.app

Group NEXA · Universiti Teknologi PETRONAS
SEDIC@FOI2026